

Probability and Conditional Probability

MGMT 670: Business Analytics



Becoming CEOs

1% of Purdue DSB MS become CEO. Among these CEOs, 90% got A+ in MGMT 670. Among the non-CEOs, 90% did not get A+ in MGMT 670.

You get an “A+” in MGMT 670, what is your chance of becoming a CEO?

- a. Higher than 50%
- b. Between 10% and 50%
- c. Between 5% than 9.99%
- d. Lower than 5%

This lecture

- Probability
- Cross-tabulation and Contingency Tables
- Conditional Probability, Multiplication law
- Bayes’ Theorem and Information update

Cross-tabulation and Contingency Tables

A BestBuy store wants to analyze customer consumption behavior. The store randomly chooses 100 recent records to obtain the following contingency table (Pivot table):

TO GET THIS FROM DATA:

Minitab: Stat -> Tables -> Cross Tabulation and Chi-Squared.....

	TV			
DVD	Regular	Flat Screen	HDTV	Total
No	20	10	10	40
Yes	5	25	30	60
Total	25	35	40	100

Questions

We use this data to calibrate

1. What is the probability that a customer purchased a DVD player **and** HDTV?
2. What is the probability that a customer purchased a flat-screen TV?
3. What is the probability that a customer purchased an HDTV **or** a DVD player?

TV and DVD Purchases

	TV			
DVD	Regular	Flat Screen	HDTV	Total
No	20	10	10	40
Yes	5	25	30	60
Total	25	35	40	100

- Buy a DVD player **and** a HDTV: _____;
- Buy a flat screen TV: _____;
- Buy a DVD player **or** HDTV: _____.

Joint Probability Using Contingency Table

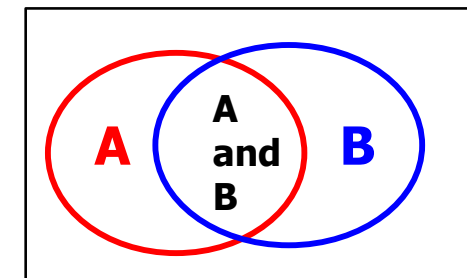
Event	Event		Total
	A_1	A_2	
B_1	$P(A_1 \text{ and } B_1)$	$P(A_2 \text{ and } B_1)$	$P(B_1)$
B_2	$P(A_1 \text{ and } B_2)$	$P(A_2 \text{ and } B_2)$	$P(B_2)$
Total	$P(A_1)$	$P(A_2)$	

Joint Probability

Marginal Probability

Venn diagram

- $\Pr(A)$, $\Pr(B)$



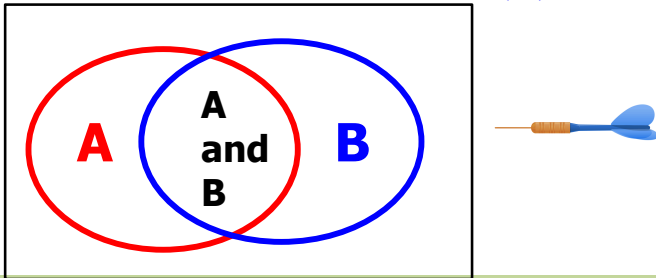
- $\Pr(A \text{ and } B)$

- $\Pr(A \text{ or } B)$

Conditional Probability

- $\Pr(B|A)$ = the conditional prob of B given A or chance of B when A occurs

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$



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	TV			
DVD	Regular	Flat Screen	HDTV	Total
No	.20	.10	.10	.40
Yes	.05	.25	.30	.60
Total	.25	.35	.40	1.00

$\Pr(\text{not DVD} | \text{Regular}) =$

$\Pr(\text{Regular} | \text{DVD}) = ?$

Multiplication Law

- The **multiplication law** provides a way to compute the probability of an intersection of two events.
- The law is written as:

$$P(A \text{ and } B) = P(B)P(A|B) \\ = P(A)P(B|A)$$

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	TV			
DVD	Regular	Flat Screen	HDTV	Total
No	.20	.10	.10	.40
Yes	.05	.25	.30	.60
Total	.25	.35	.40	1.00

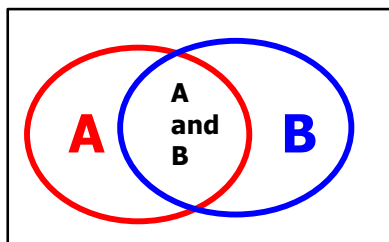
$\Pr(\text{DVD} | \text{HDTV}) =$

$\Pr(\text{DVD} | \text{FlatScreen}) =$

Becoming CEOs

1% of Purdue DSB MS become CEO. Among these CEOs, 90% got A+ in MGMT 670. Among the non-CEOs, 90% did not get A+.

You get an “A+” in MGMT 670, what is your chance of becoming a CEO?



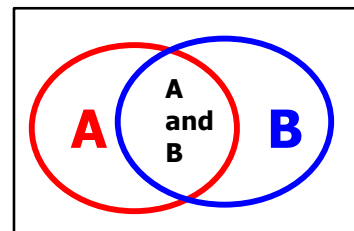
A= getting A+
B= becoming CEO

Need to calculate
 $\Pr(A|B)$ or $\Pr(B|A)$?

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	A+	Not A+	total
CEO			
Not CEO			
total			



A= getting A+
B= becoming CEO

$\Pr(\quad | \quad) =$

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Becoming CEOs

1% of Purdue DSB MS become CEO. Getting an “A+” in MGMT 670 **has no implication** for a Purdue MS student to become a CEO in the future.

4% of students gets “A+”. You got A+, **what is your chance of becoming a CEO?**

- a. 1%
- b. 4%
- c. 0.4%
- d. 0.04%

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Becoming CEOs

1% of Purdue DSB MS become CEO. Getting an “A+” in MGMT 670 **has no implication** for a Purdue MS student to become a CEO in the future.

4% of students gets “A+”. **What is your chance of getting A+ and becoming a CEO?**

- a. 5%
- b. 4%
- c. 0.4%
- d. 0.04%

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Statistically Independent Events

- Getting A+ has nothing to do/ independent with becoming CEO means:

$$\Pr(\text{CEO} | A+) = \Pr(\text{CEO})$$

$$\Pr(\text{CEO and } A+)/\Pr(A+) = \Pr(\text{CEO})$$

Thus

$$\Pr(\text{CEO and } A+) = \Pr(A+) \times \Pr(\text{CEO})$$

- Events A and B are said to be **statistically independent** if knowing one event happened or not does not affect the probability of the other event happening or not.
- A and B are statistically independent if any of the following conditions is satisfied:

$$P(A | B) = P(A),$$

$$P(B | A) = P(B),$$

$$P(A \text{ and } B) = P(A) \times P(B).$$

Value of information

Perfect information
(fully correlated)



Collect information

Not all information has
the same additional value

Independence,
or no information

TECH 2/16/2012 @ 11:02AM | 2,497,967 views

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

+ Comment Now + Follow Comments

<http://www.forbes.com/>

Every time you go shopping, you share intimate details about your consumption patterns with retailers. And many of those retailers are studying those details to figure out what you like, what you need, and which coupons are most likely to make you happy. Target, for example, has figured out how to data-mine its way into your womb, to figure out whether you have a baby on the way long before you need to start buying diapers.



Target has got you in its aim

Charles Duhigg outlines in the New York Times how Target tries to hook parents-to-be at that crucial moment before they turn into rampant — and loyal — buyers of all things pastel, plastic, and miniature. He talked to Target statistician Andrew Pole — before Target freaked out and cut off all communications — about the clues to a customer's impending bundle of joy. Target assigns every customer a Guest ID number, tied to their credit card, name, or email

Updating Prior Probabilities Using Sample Information

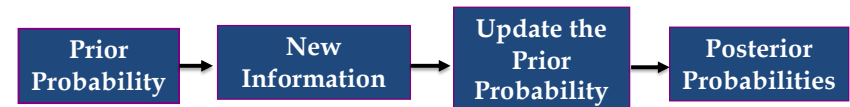
- Examples:
 - Dynamic Pricing: decide the price of a new product, based on info from other related products, marketing environment, etc.
 - Marketing: Evaluating success probability of a product in the real market, based on results from a test market.
 - Recommendation Systems: personalize recommendation based on a user's past behavior.

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Bayes' Theorem

- Often we begin probability analysis with initial or prior probabilities (from subjective assessment).
- Then, from a sample, special report, or a product test (experiment) we obtain some additional (sample) information.
- Given this information, we revise the prior probabilities to obtain posterior probabilities.



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Bayes' Theorem

- **Formula:**

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n)}$$

$\underbrace{P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B)}_{P(B)}$

- Bayes' theorem is applicable when the events for which we want to compute posterior probabilities are mutually exclusive and collectively exhaustive.

Example: Test Marketing

- A new banking account product can be:
 - *F*: a *failure* in the real market or
 - *M*: a *marginal success* in the real market or
 - *S*: a *major success* in the real market

Prior Probabilities

- It is known, from the past experience with similar products or based on the management's judgment, that

$$P(F) = 0.80, \quad P(M) = 0.15, \quad P(S) = 0.05$$

- These probabilities are called *prior probabilities*.

Case Study: Test Market

- Before introducing the product to the real market, the company has an option to test it in a test market.
- Possible outcomes in the test market are:
 - *Small*: a *small* test market share or
 - *Large*: a *large* test market share.

Historical Information Shows ...

- 2% of all products that were failures in the real market (F) had a large test market share (Large);
- 24% of all products that were marginal successes in the real market (M) had a large test market share (Large); and
- 98% of all products that were major successes in the real market (S) had a large test market share (Large).

Management's Question

- Marketing wishes to know the probabilities of ultimate success of the product, given different possible test market results. That is, to find

$$P(S | \text{Small}) \text{ and } P(S | \text{Large}) .$$

- These are called *posterior probabilities*.

	Market Test result		
	Small	Large	Total
F:failure			
M: Marginal			
S:success			
Total			

$P(S | \text{Small}) = ?$

$P(S | \text{Large}) = ?$

Do you think this market test is useful?

Summary

- Probability
- Cross-tabulation and Contingency Tables
- Conditional Probability, Multiplication law
- Bayes' Theorem and Information update